

CY 501P: Organic Chemistry Laboratory

Course Code: CY 501P

Course Name: Organic Chemistry Laboratory

L-T-P-C: 0-0-8-4

Prerequisites: Core course

Intended for: M.Sc in Chemistry

Distribution: Core course

Approval: 9th Senate

Course Contents

- Techniques [3 Lectures]
- Learning structure elucidation of organic molecules & determination of following functional groups using IR and NMR spectroscopy-: (a) Amines (b) Imines (c) Esters (d) Amides (e) Carboxylic acid (f) Alcohol (g) Ethers (h) Aldehydes/Ketones [3 Lectures]
- Extraction of Organic compounds & their characterization using IR, NMR and Mass spectroscopy-: (a) Extraction of caffeine from tea leaves. (i) Extraction of piperene from black pepper powder. [6 Lectures]
- Quantitative Estimations-: Estimation of carbonyl group-: (i) Standardization of glucose solution by Fehlings Method. (ii) Determination of percentage purity of given sample of glucose. [3 Lectures]
- Organic Synthesis-: Synthesis of organic molecules & their characterization using IR, NMR and Mass spectroscopy. (a) One step synthesis-: Synthesis of Bezpinaol (b) Multistep Synthesis-: Synthesis of Benzilic Acid; Benzaldehyde Benzoin Benzil Benzilic acid [6 Lectures]
- Synthesis of a fluorogenic molecule and evaluation of its quantum yield. [4 Lectures]
(a) Introduction to c-c couplings. (b) Significance of c-c couplings in pharmaceutical industry. (c) Performing Sonogashira coupling at laboratory scale. [8 Lectures]
- Synthesis of organic reagent, use of reagent in organic reaction & determination of its optical activity. (a) Synthesis of N-Bromosuccinamide (organic reagent, involves highly toxic and fuming Br₂ for synthesis). (b) Use of NBS in synthesis of halohydrin (Electrophilic Addition Reaction). (c) Determination of optical activity of generated product using polarimeter. [7 Lectures]
- Use of Computer - Chem Draw, Draw the structure of simple aliphatic, aromatic, heterocyclic compounds with different substituents. Get the correct IUPAC name and predict the ¹H NMR signals. [1 Lectures]
- Practical demonstration of instruments (a) HPLC (b) NMR [1 Lectures]

NOTE- Alternate practical may be offered subject to conditions and availability of chemicals.

Books Recommended

1. **Vogels textbook of practical organic chemistry**, John Wiley and Sons.
2. **Vogels elementary quantitative organic analysis**, John Wiley and Sons.
3. Robert, Gilbert & Rodewald, **An Introduction to Modern Experimental Organic Chemistry**, Saunders Publication.



4. Skoog, West and Holler, **Fundamentals of Analytical Chemistry**, Saunders College publishing.
5. Field, Sternhell & Kalman, **Organic structures from Spectra**, John Wiley and Sons.